

More people. More exposure. Fewer storm deaths per person.

U.S. fatalities from tornadoes, floods and hurricanes have fallen sharply relative to population since the 1940s, even as migration has added people to exposed coastal counties. Evidence of adaptation, not immunity.

Michael Scholz · University of Rochester · Fall 2026

18.8

annual deaths per 10 million
in the 1940s

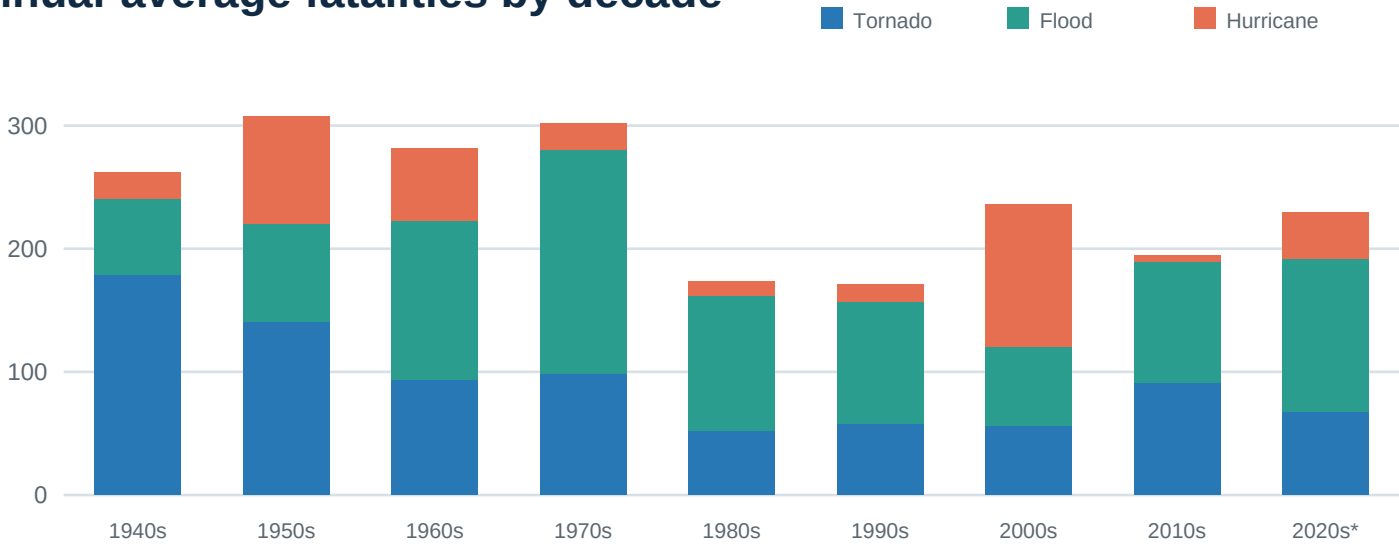
6.8

annual deaths per 10 million
in 2020–2025

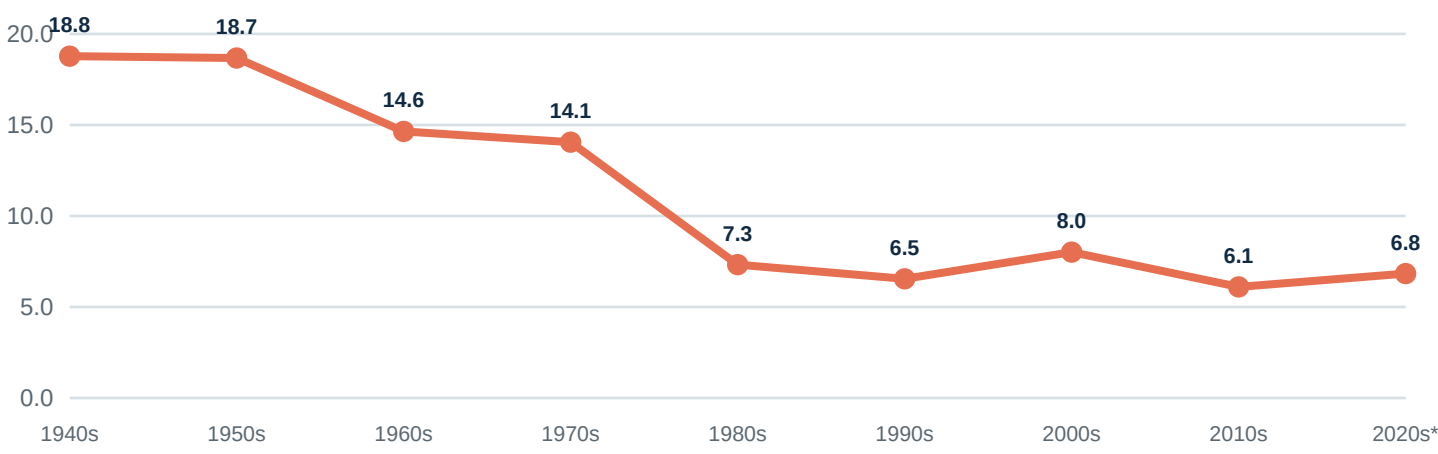
64%

lower per-person
rate

Annual-average fatalities by decade



Annual deaths per 10 million Americans



ECONOMICS OF RISK

Risk = hazard × exposure × vulnerability

Warnings, shelters and stronger buildings can lower vulnerability. Land-use rules and relocation can lower exposure. The efficient mix depends on marginal lives saved per dollar and on who bears the costs.

WHY PEOPLE STILL MOVE TOWARD RISK

Risk is only one part of the choice

People also value jobs, housing, amenities, family and warm weather. A risky move can be rational. Subsidized insurance or disaster aid, however, can shift expected costs to others.

MIGRATION AND HURRICANE EXPOSURE

Some domestic migration has moved toward the South Atlantic coast

2016–2018 Brunswick County, NC and Horry County, SC had high domestic-migration rates; Brunswick ranked first nationally in 2018.

2025 Jasper County, SC, near Hilton Head, reached the top domestic-migration spot.

What it adds: people moved into exposed coastal counties, not merely southern states. But a coastal address is not a precise hurricane-risk measure, and the migration evidence does not identify the cause of the mortality trend.

A measurement trap: ‘heat deaths’ can mean two different things

Hundreds

NWS counts deaths directly reported under a heat hazard category.

≈120,000

Burke et al. estimate annual U.S. deaths statistically attributable to nonoptimal temperature.

The Burke estimate uses changes in all-cause mortality across temperatures; it is not a count of death certificates labeled heat or cold. The paper estimates about 5% of U.S. deaths are temperature-attributable and finds cold causes more deaths than heat. Both measures are useful, but combining them would be misleading.

BUT WAIT

Fewer deaths are not the whole welfare calculation

Mortality can fall while property losses, displacement or unequal burdens rise. Reporting changed; extreme years matter; NWS hurricane counts are not total tropical-cyclone deaths; and the 2020s are incomplete.

THE IPCC CONTEXT

Exposure drives disaster losses

SREX concludes with high confidence that increasing exposure of people and assets is the major cause of long-run increases in weather- and climate-disaster losses. Dollar losses are not fatalities, but where and how we build changes damage.

The question economics adds Which combination of warnings, building standards, infrastructure, insurance pricing and land-use choices reduces risk at the lowest social cost without hiding who pays and who remains exposed?



Sources: National Weather Service, 81-Year List of Severe Weather Fatalities (1940–2025); U.S. Census Bureau historical population estimates and County Domestic Migration Trends (2011–2025); Burke et al. (2025), Understanding and Addressing Temperature Impacts on Mortality; IPCC SREX (2012). Full methods and live links on the webpage. 2025 NWS data preliminary.